

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-10 18:00 UTC

SEC P vs NP — notebook summary

Generated 2026-05-10 18:00 UTC

- Cycles recorded: **557**
- Time span: 404.3h (~1.38 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	520
BARRIER_HIT	18
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	30
Representation Theory of Symmetric Groups	17
Free Probability	14
Schur-Weyl Duality	14
Matroid Theory	11
Additive Combinatorics	8
Algebraic Geometry	7
Polymatroid Theory	7
Random Matrix Theory	6
Finite Geometry	6
Spectral Graph Theory	6
Plethysm Theory	6
Noncommutative L^p Geometry	5
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4
Algebraic Geometry of Secant Varieties	4
FOURIER_ANALYSIS	4
Fourier analysis of boolean functions	3

Field	Cycles
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3
Schur-Weyl duality	3
Fourier Analysis of Boolean Functions	3
Persistent Homology	3
Noncommutative Geometry	3
Noncommutative geometry	2
Representation theory of symmetric groups	2

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-10 12:05 UTC	INCONCLUSIVE	Matroid Rank Inverse Proportional to Monotone DNF Discrepancy for
2026-05-10 12:27 UTC	INCONCLUSIVE	Symmetric Group Fourier Coefficient Sum Exponential Gap Between P
2026-05-10 12:38 UTC	INCONCLUSIVE	Schur-Weyl Duality Invariant for Permanent vs Determinant Circuit
2026-05-10 12:46 UTC	INCONCLUSIVE	Schur-Weyl Dimension Gap in 3-CNF Permanent vs Determinant Shapes
2026-05-10 13:07 UTC	INCONCLUSIVE	Noncommutative L^p Norm Lower Bound for Disjointness Communicatio
2026-05-10 13:12 UTC	BARRIER_HIT	Schur Coefficient Gap in Permanent vs Determinant Decompositions
2026-05-10 14:15 UTC	INCONCLUSIVE	Secant Variety Dimension Gap in Determinant vs Permanent Orbits f
2026-05-10 14:31 UTC	INCONCLUSIVE	Fourier Coefficient Discrepancy and ACC^0 Circuit Size
2026-05-10 14:39 UTC	INCONCLUSIVE	Fourier Coefficient Concentration and ACC^0 Circuit Size
2026-05-10 14:54 UTC	INCONCLUSIVE	Moment Matrix Rank Lower Bound for Max-CUT SOS Approximation
2026-05-10 15:10 UTC	INCONCLUSIVE	Kronecker Coefficient Non-Zero Threshold for Permanent-Complete S

Time	Verdict	Title
2026-05-10 16:28 UTC	INCONCLUSIVE	SOS Degree Lower Bound via Matroid Polytope Dimension
2026-05-10 16:34 UTC	BARRIER_HIT	Schur Coefficient Exponential Gap in 3-CNF Permanent vs Determina
2026-05-10 17:22 UTC	INCONCLUSIVE	Symmetric Group Fourier Coefficient Gap in Disjointness Communica
2026-05-10 17:37 UTC	INCONCLUSIVE	Harmonic Coefficient Sum Lower Bound for Resolution Proof Length

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```